

Project SAFEMARVEL

(A **S**afety-critical control **F**ramEwork for roll stabilization of **M**ARine **V**esseLs)



DACOMA



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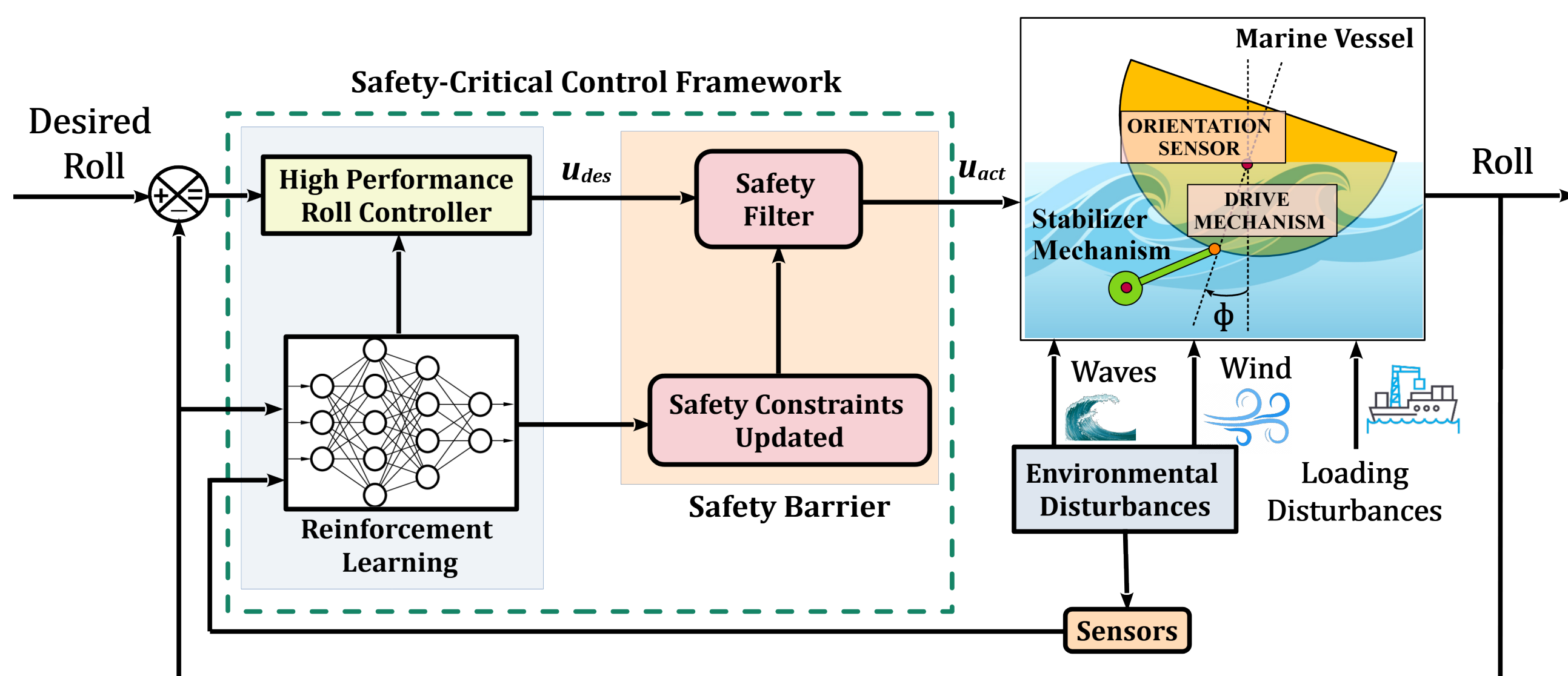
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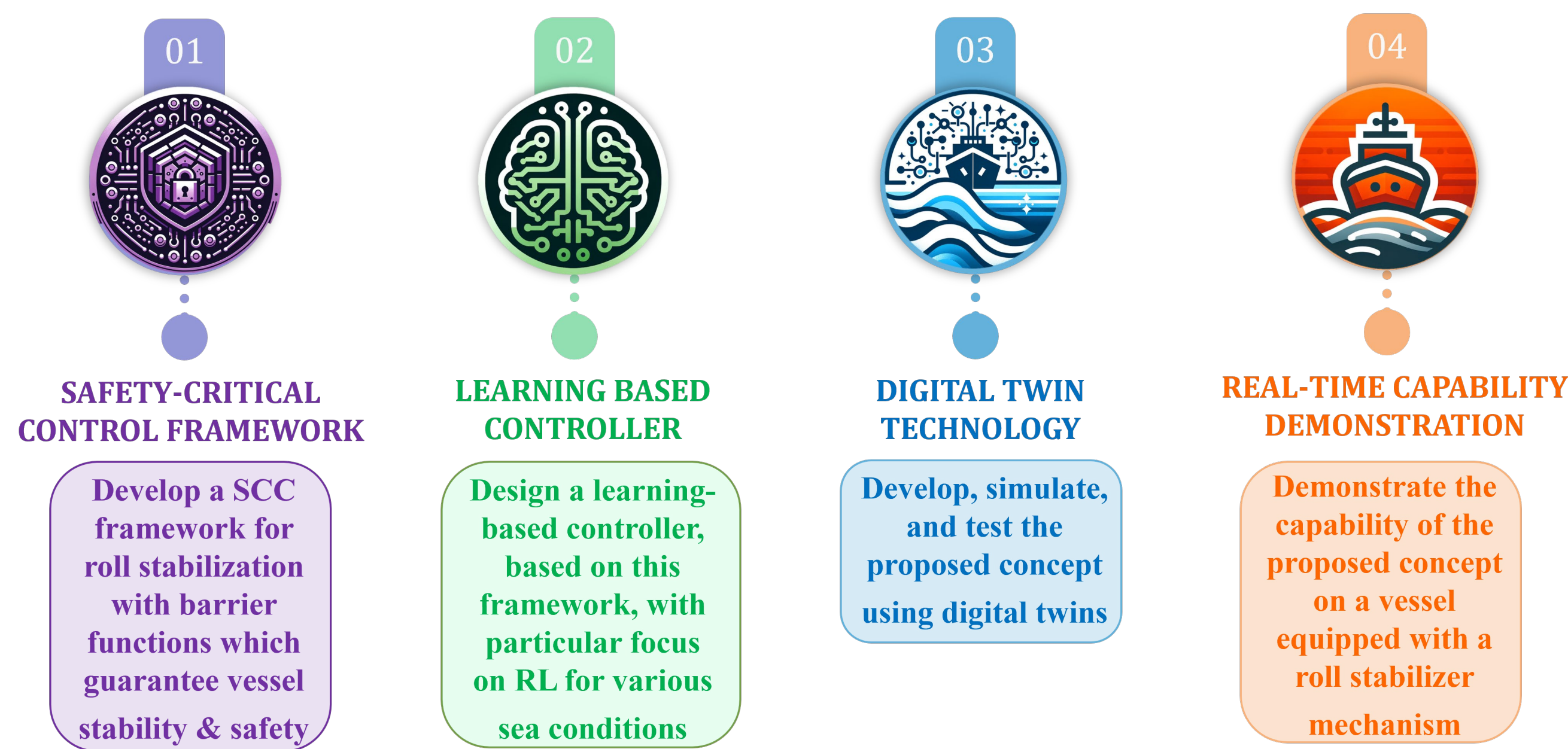
Project Background

Project SAFEMARVEL began in March 2024 and aims to develop a safety-critical control framework for roll stabilization of marine vessels, ensuring vessel safety through the use of barrier functions. The project integrates high-performance, learning-based controllers adapted to varying sea conditions. Using digital twins, the framework will be tested in simulation and demonstrated in real-time on a demo vessel. Led by SDU Mechatronics in collaboration with Dacoma ApS and SDU Physics, the project brings together academic and industrial partners to advance safety solutions for the Danish maritime sector. The project will end in February 2026.

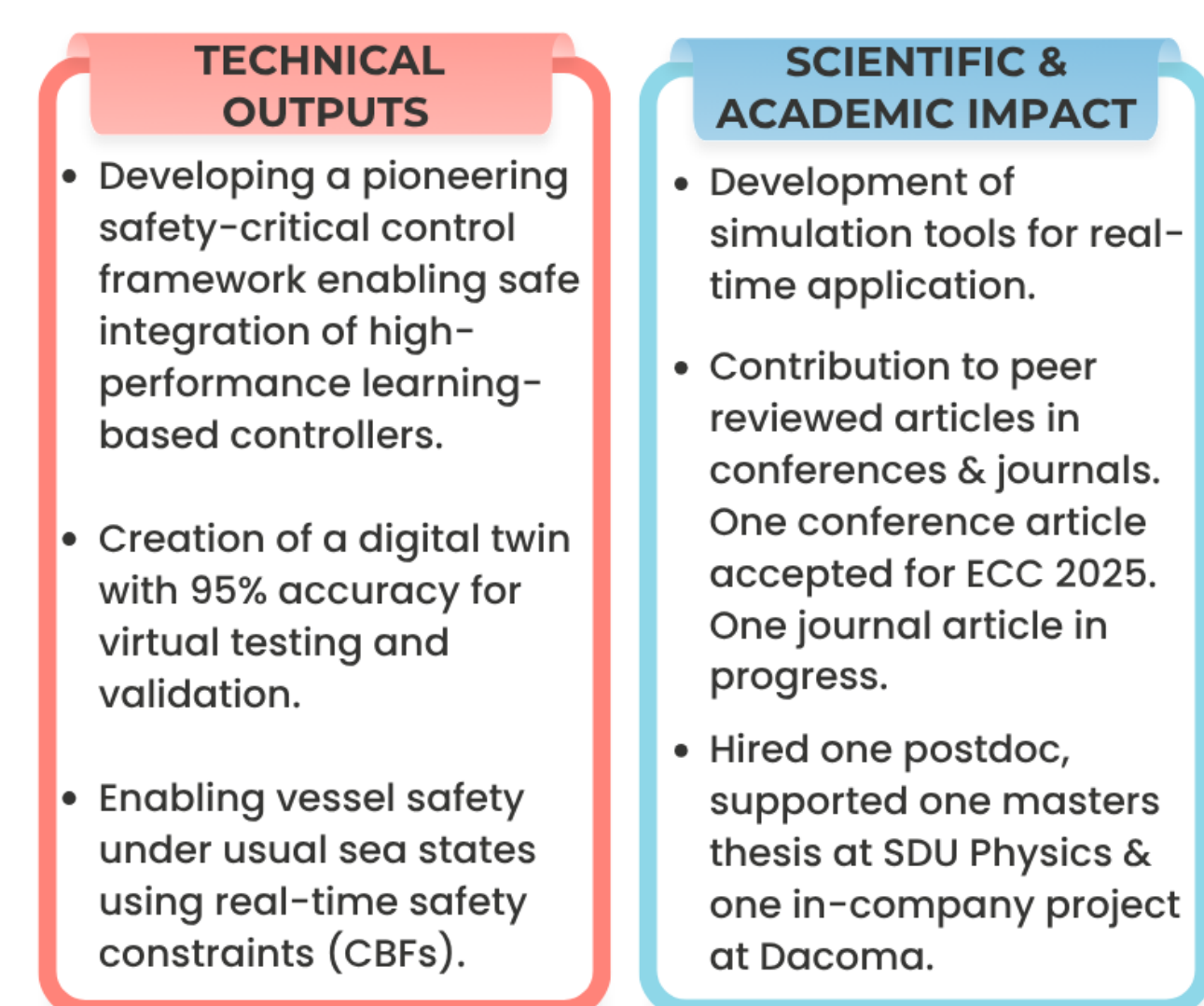


Schematic of the proposed safety-based learning framework for vessel stabilizers

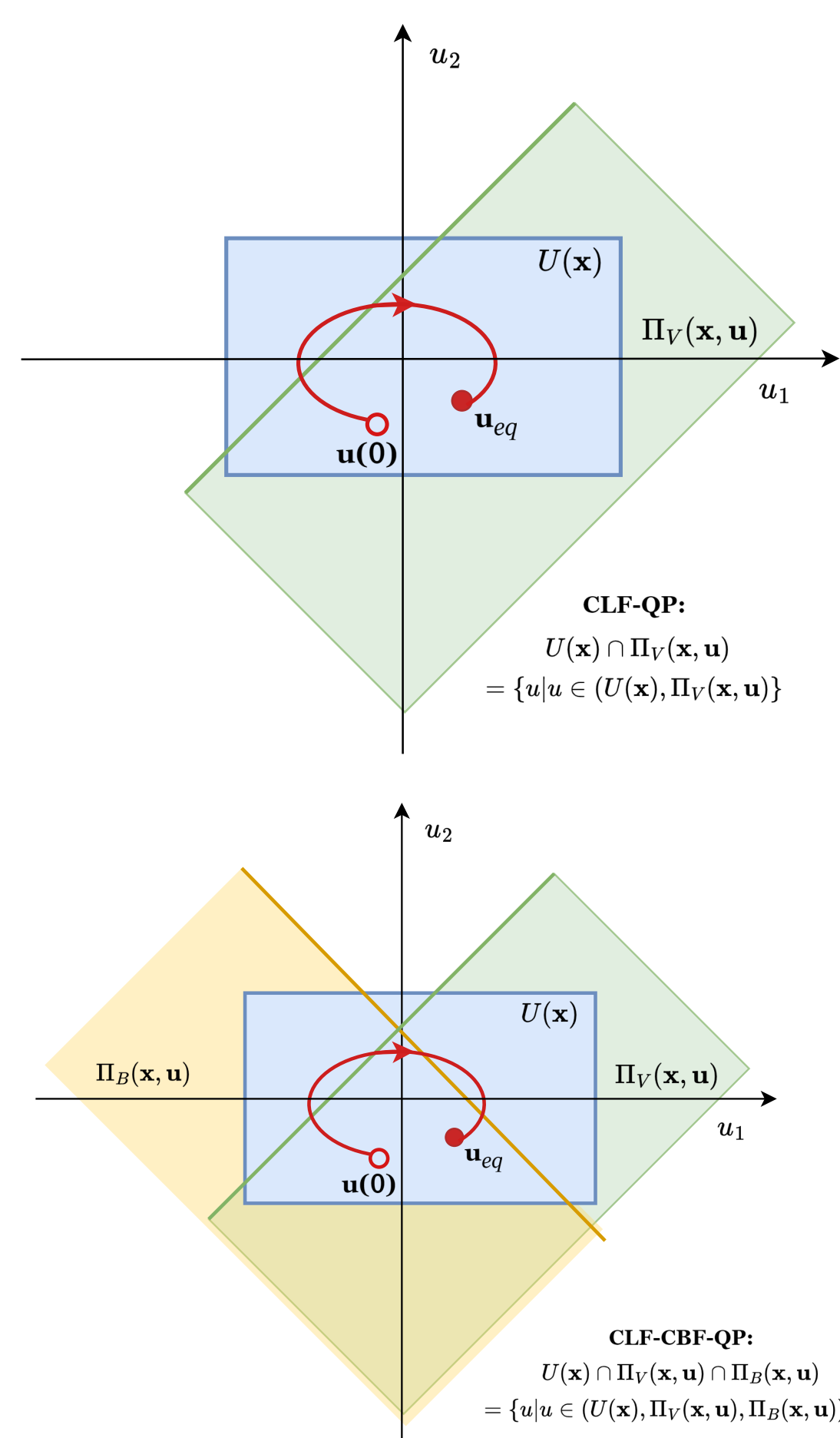
Project Objectives



Project Outcomes (Planned)



Snapshot from bi-annual meeting of the SAFEMARVEL team at SDU Sønderborg



Schematic illustrating the intersection of the constraints for the proposed QP formulations and the trajectory of the control solution offered by the nominal controller*

Project AMCOSTAR

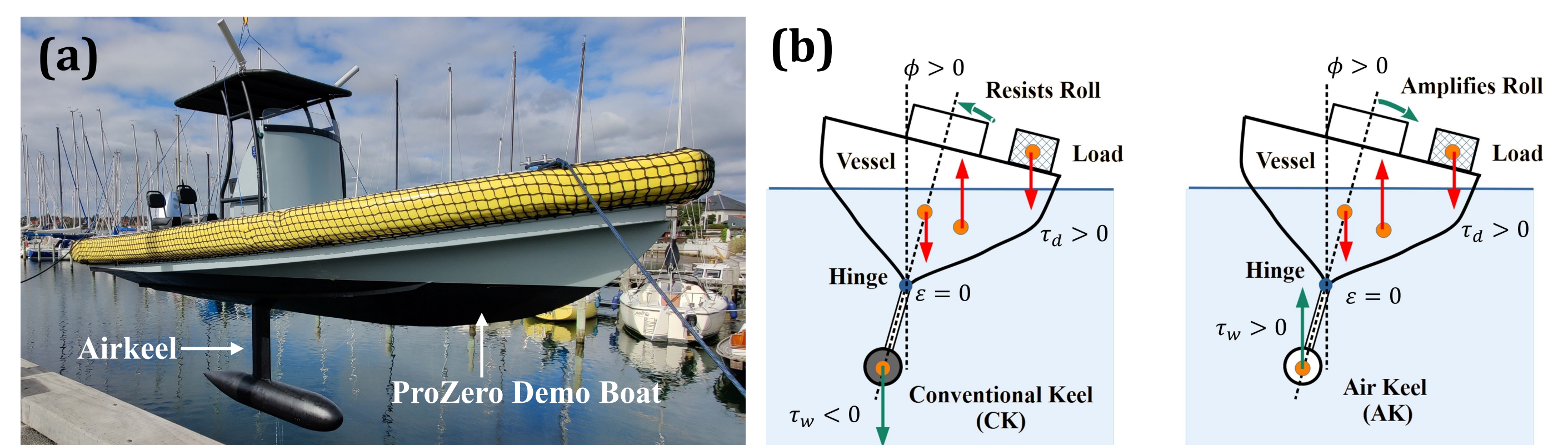
(**D**ynAMic Airkeel Stabilizer for Increasing the **C**apacity Of **S**maller **B**oaTs through **A**ctive **C**ontrol)

Innovation Fund Denmark



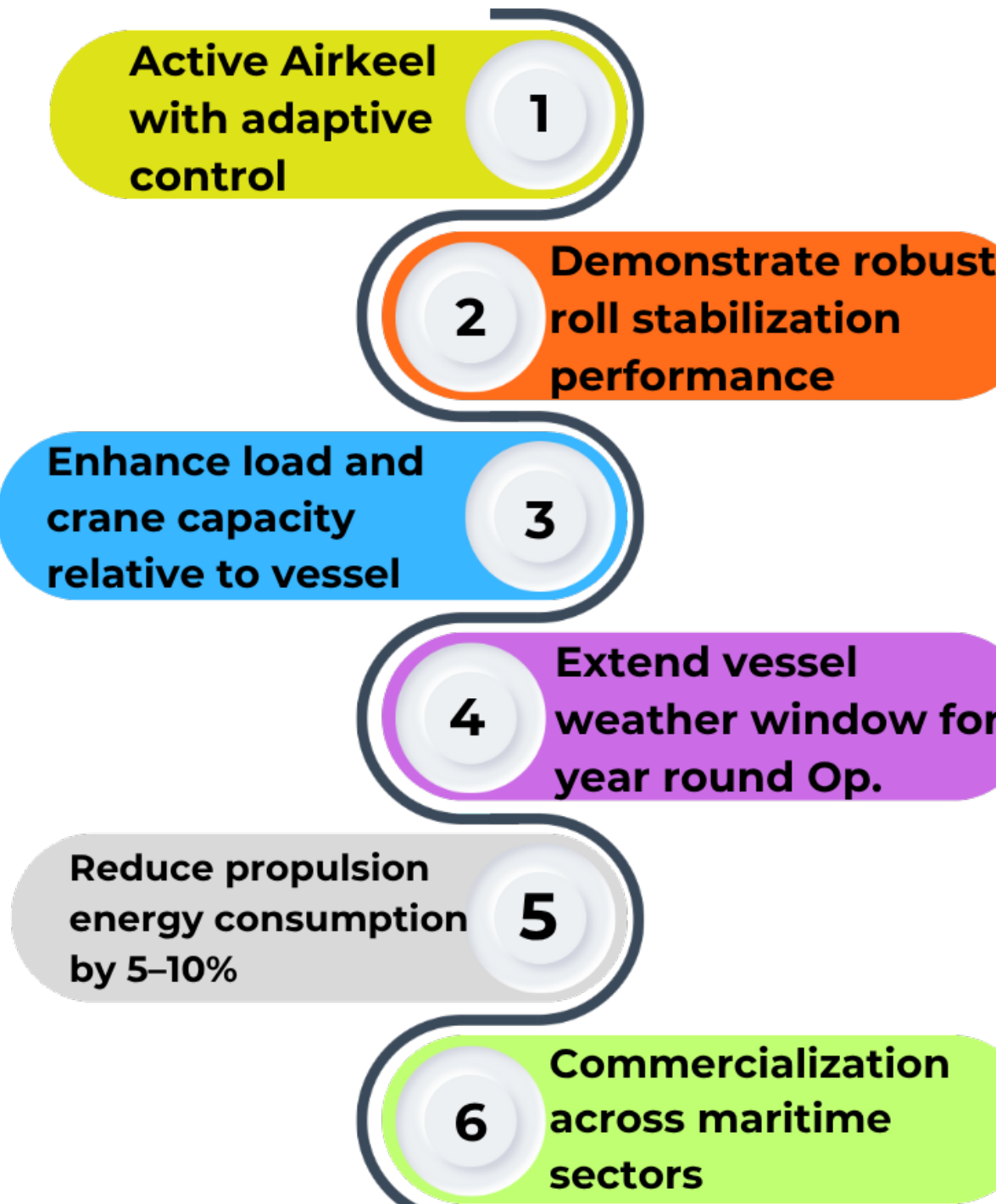
Project Background

The AMCOSTAR project, a Eurostars/Eureka and Innovation Fund Denmark initiative, was completed in July 2023. It aimed to develop and demonstrate an integrated air-filled canting keel (dubbed as *Airkeel*) and adaptive control system (ACS) to enhance the stability, capacity, and energy efficiency of small to medium-sized vessels (6–40 meters). By transforming passive stabilization into an intelligent, responsive system, the project sought to improve safety, increase load and crane capacity, reduce fuel consumption, and expand operational capabilities, particularly in offshore wind and aquaculture sectors. Through digital twin modeling, sea trials, and close collaboration within the consortium comprising Dacoma ApS, TUCO Group, SDU, and Automasjon & Data, AMCOSTAR delivered a viable stabilization system tailored to the needs of aquaculture vessels and CTVs.

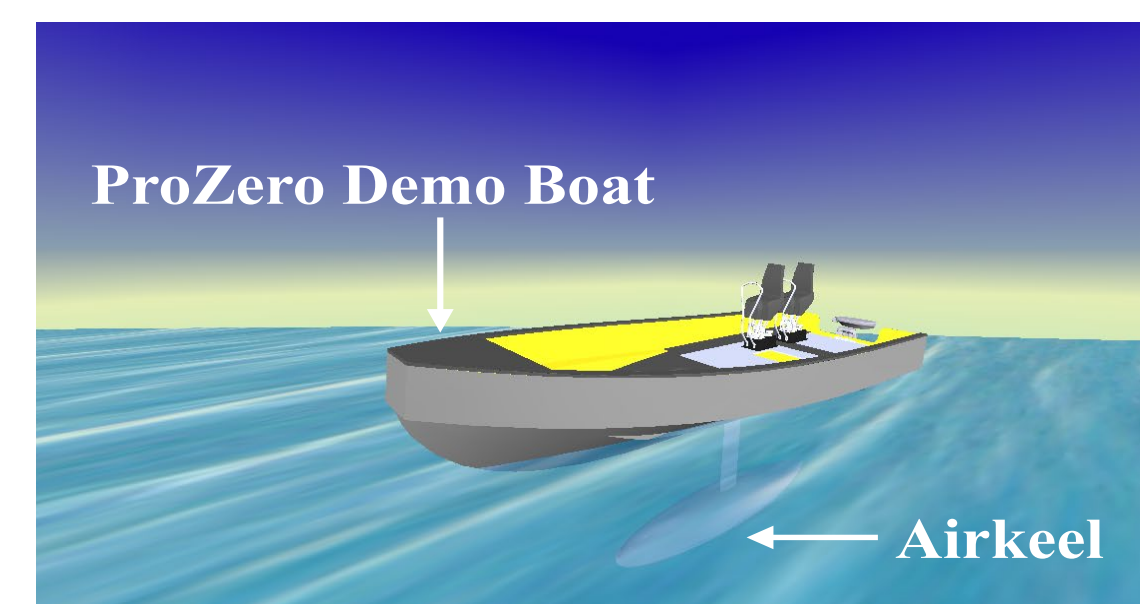
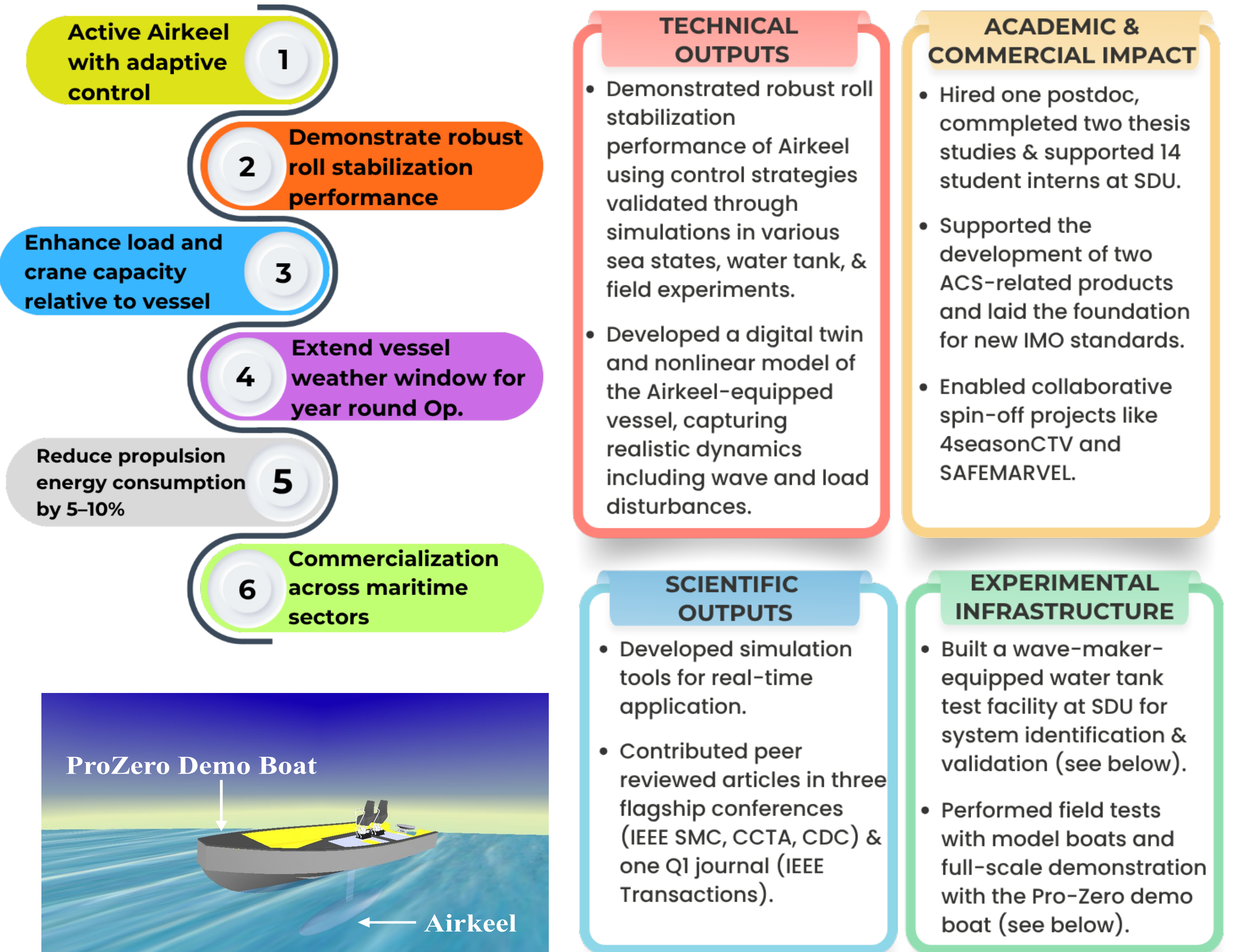


(a) Photographic view of the ProZero Demo boat equipped with the airkeel mechanism
(b) Schematic representation of the effect of load disturbance on the vessel with CK and AK **

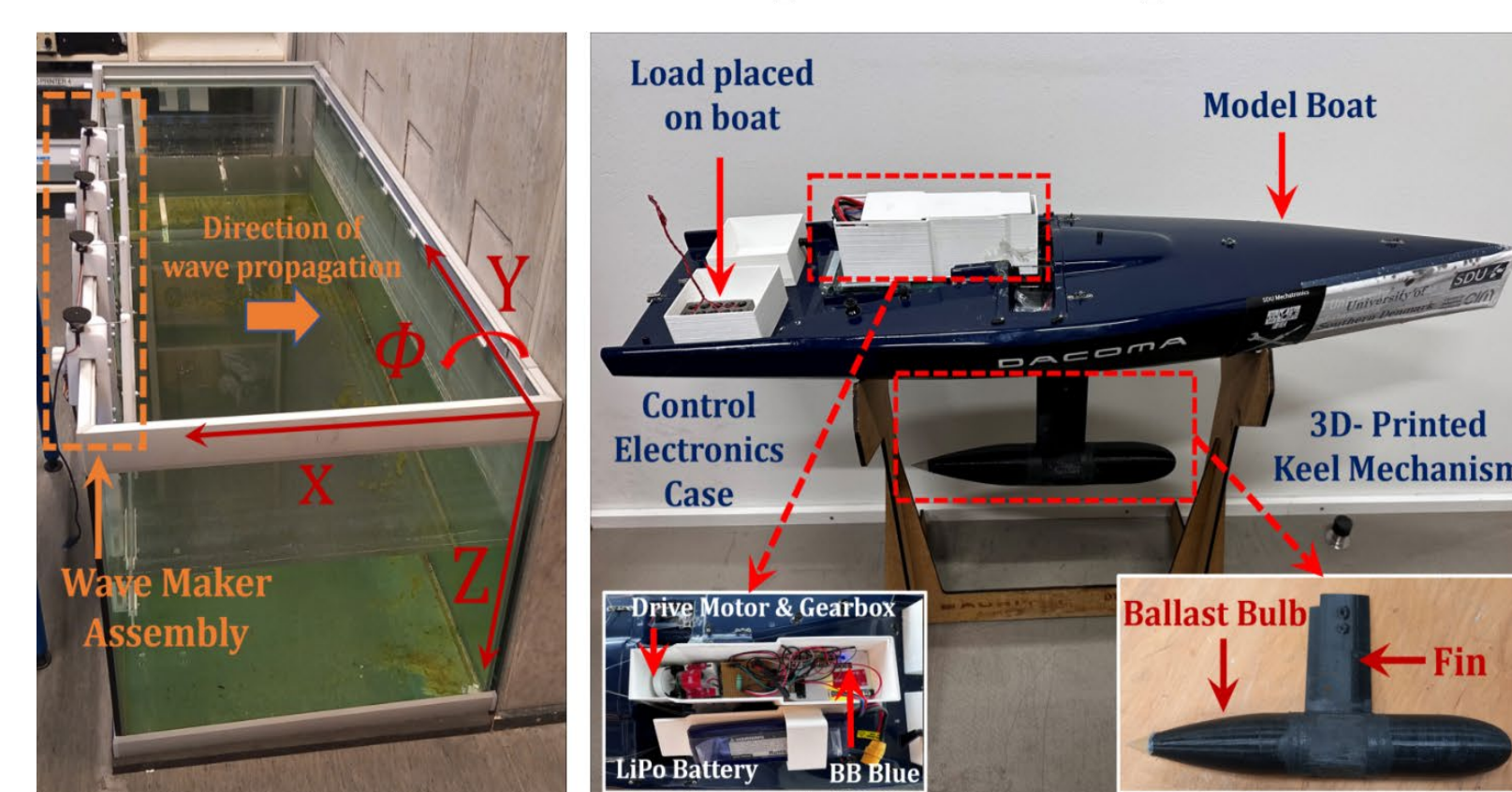
Project Objectives



Project Outcomes

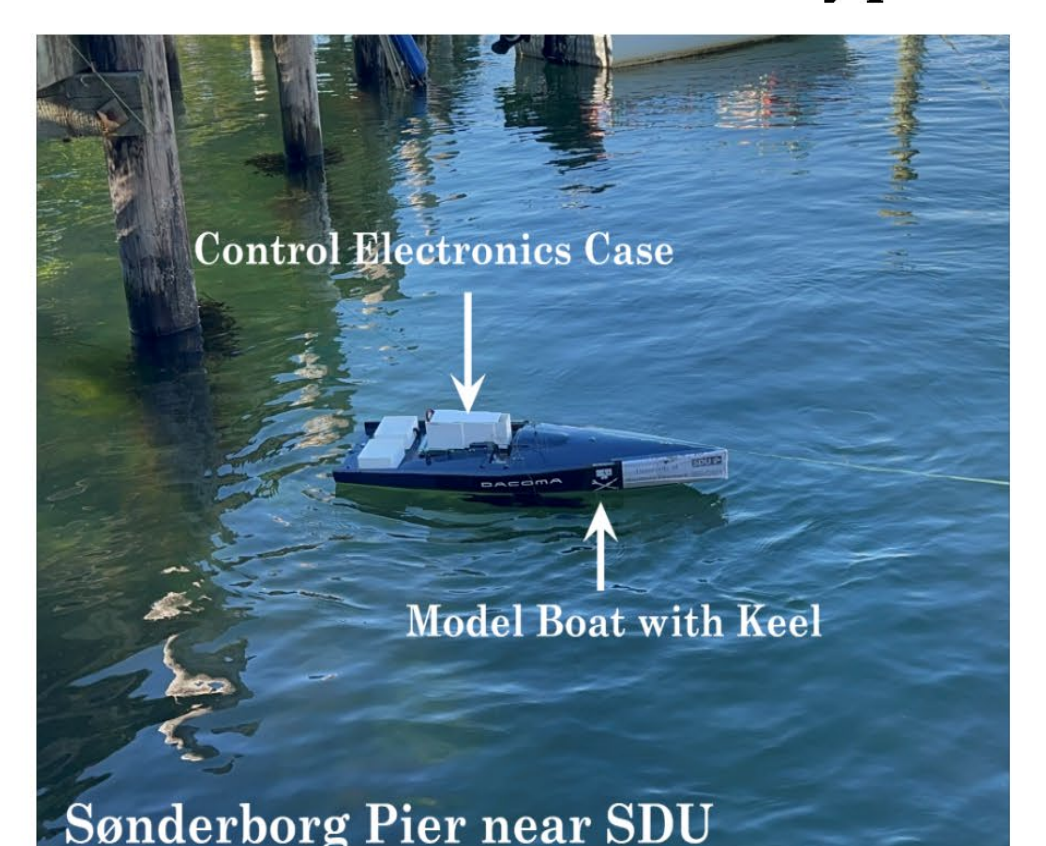


1. Water Tank Setup for Indoor Experiments



(a) Water Tank Setup (b) Model Boat for Testing with Embedded Systems

2. Field test location at nearby pier



Model Boat during Field Test for wave disturbances

(1) Layout of the indoor experimental setup (2) Snapshot of the field test with the model boat in a stream of water near the Sønderborg pier (Denmark) **

*S. Chaudhuri, H. Ramezani, and J. Jouffroy, "Remarks on augmenting stability and safety in nominal controllers: a practical perspective," accepted for presentation at European Control Conference 2025, 24 -27 June.

**H. Ramezani, S. Chaudhuri, J. Jouffroy, A. Baurichter and S. Matrup Hansen, "On Roll Stabilization Using a Canting Keel," in IEEE Transactions on Control Systems Technology, vol. 33, no. 2, pp. 539-553, March 2025, doi: 10.1109/TCST.2024.3494237.